IJAY NARANG

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EDUCATION

Georgia Institute of Technology | Atlanta, GA | GPA 3.9/4.0 *2025-Present*
- M.S. in Machine Learning

Princeton University | Princeton, NJ | GPA 3.9/4.0 *2021-2025*
- B.S.E. in Computer Science, summa cum laude, with minors in Applied Mathematics, Finance, and Optimization

SELECTED PUBLICATIONS & PREPRINTS

(ICML 2026) **Steady-State Behavior of Constant-Stepsize Stochastic Approximation: Gaussian Approximation and Tail Bounds** (with Zedong Wang, Yuyang Wang, Felix Wang, Yuzhou Wang, and Siva Theja Maguluri) [View Paper Here]

(SODA 2027) **Computational Thresholds for Balanced and Fixed-Slice Independent Sets in Bipartite Graphs** (with Will Perkins, Yuzhou Wang, and Timothy L. H. Wee) [View Paper Here]

(Submitted to CPC) **Structural Corrections to the Bethe Approximation of the Permanent** (with Will Perkins) [View Paper Here]

(Submitted to ALT 2027) **Optimal Detection of Planted Stars via a Random Energy Model** (with Will Perkins and Timothy Wee) [View Paper Here]

(Preprint) **The Hard-Core Model on Bipartite Spectral Expanders: Counting and Sampling at All Fugacities** (with Will Perkins) [View Paper Here]

(Manuscript) **On even-H-free Colorings** (with Noga Alon) [View Paper Here]

WORK EXPERIENCE

Princeton Vision & Learning Lab | *Visiting Researcher* *Summer 2026*
- Worked with Jia Deng on GlimpseBC, a behavioral cloning algorithm that learns saccadic visual policies from full-frame demonstrations. The resulting policies achieve competitive performance with full-frame baselines on RoboMimic and Push-T while reducing computational cost. Paper in progress.

Capula Investment Management | *Trading and Research Intern* *Summer 2024*
- Built a monitoring system for correlations and dislocations between CDS spreads. Developed a backtester and used it to evaluate and optimize relative-value trading strategies.

Siemens Digital Industries | *Software Development Intern* *Summers 2022 and 2023*
- Implemented frontend components displaying client policies for the Operations Insights Dashboard with React.js.
- Developed an invoice generator using PostgreSQL for the Digital E-Commerce billing module.

Solar Operations Solutions, LLC | *Machine Learning Intern* *Summer 2021*
- Used linear time-series models to forecast energy demand and regulate supply across solar sites.

AWARDS AND HONORS

- Georgia Tech Presidential Fellowship (2026)
- Computer Science Outstanding Student Teaching Award, Princeton University (2025)
- Outstanding Computer Science Research Prize, Princeton University (2025)
- Computer Science Service Award, Princeton University (2025)
- Highest Honors, Princeton Computer Science Department (2025)
- 4th Place at ICPC Greater New York Regional Competition (2023)
- U.S. Presidential Scholar (2021)

SKILLS AND PROJECTS

- **Languages and Libraries:** Python, OCaml, R, Java, SQL, PyTorch, NumPy, SciPy, pandas, and scikit-learn
- **Compass:** Built an all-in-one Princeton University course-planning app, now used by 700+ students and maintained by Hoagie Club as Hoagie Plan. [See the App Here]
- **Chess:** Achieved a Second Category title with a peak USCF rating of 1902.